USN | | | | | | | | | | | [24MTEVT101]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

ELECTRIC AND HYBRID VEHICLES

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Discuss the evolution of electric vehicles (EVs) from the past to the present.	CO1	L3	10
	b.	Compare the technological advancements of state-of-the-art electric vehicles (EVs) and hybrid electric vehicles (HEVs) currently available.	CO1	L2	10
2.	**a.**	Discuss the factors influencing the choice of an electric propulsion system in electric vehicles (EVs).	CO2	L3	10
	b.	Discuss the different types of power electronics converters used in EV applications. Highlight their specific roles in motor control and battery charging.	CO2	L2	10
3.	**a.**	Describe the role of a DC-DC converter in DC motor drives. How does it contribute to efficient speed control and energy management?	CO3	L3	10
	b.	Describe the operation of a three-phase inverter-based induction motor drive.	CO3	L2	10
		OR			
4.	**a.**	Explain the principle of Field-Oriented Control (FOC) for induction motors. How does it improve motor performance in terms of torque and speed response?	CO3	L3	10
	b.	Explain the concept of sliding mode control and its application in induction motor drives.	CO3	L2	10
5.	**a.**	Explain the differences between series, parallel, and series-parallel HEV configurations. Provide detailed diagrams for each.	CO4	L2	10
	b.	Discuss the operating principles and powertrain architecture of a series hybrid electric vehicle (HEV).	CO4	L2	10

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		OR			
6.	**a.**	Describe the power flow control mechanism in a series HEV.	**CO4**	**L2**	**10**
	b.	Explain the role of energy management systems (EMS) in optimizing power flow control in hybrid electric vehicles.	**CO4**	**L2**	**10**
7.	**a.**	Compare the different types of batteries used in electric vehicles and discuss their advantages and limitations.	**CO5**	**L2**	**10**
	b.	Explain the role of microprocessor-based charger circuits in optimizing the charging process for EV batteries.	**CO5**	**L2**	**10**
		OR			
8.	**a.**	Explain the concept of battery swapping stations and move-and-charge zones.	**CO5**	**L2**	**10**
	b.	Analyze a case study of a successful EV charging infrastructure project in major cities in India.	**CO5**	**L3**	**10**